CS 639 Homework 0: Readiness Self-Check

- **Due** Friday, September 11, on Gradescope (the exact time is on Gradescope). If you are asking to waive a prerequisite, submit by **Tuesday, September 8**. Everyone enrolled must submit HW0; it does not count toward the grade, and slip days do not apply.
- **What it is for.** HW0 is short by design. It covers the probability, linear algebra, calculus, and Python that the course assumes, and it makes sure your Python setup works. Expect about an hour. Submissions are checked for completeness, not graded for correctness. If a part does not come to you, write down where you got stuck and move on. If anything feels unfamiliar, use the pointer under each problem and ask on Piazza early (private posts are fine).
- **Do it alone.** A self-check only works if the work is yours: no classmates and no AI tools for Problems 1–4 (AI label **Red**). For installing Python and its packages, use any help you like and say so in your Assistance Statement (**Yellow**).
- **What to submit.** One file on Gradescope: `hw0.pdf`, with your answers to Problems 1–3, the output and figure from Problem 4, and the self-assessment of Problem 5. It must be typeset in LaTeX; handwritten or scanned work is not accepted. The LaTeX source of this handout is provided as a template.

1 Probability

(a) Let X be the number shown by one roll of a fair six-sided die, and let $S = X_1 + X_2$ be the sum of two independent rolls. Compute $\mathbb{E}[X]$, $\mathrm{Var}(X)$, $\mathbb{E}[S]$, and $\mathrm{Var}(S)$.

Solution.

$$\mathbb{E}[X] = \frac{7}{2}$$
$$\mathrm{Var}(X) = \frac{35}{12}$$
$$\mathbb{E}[S] = 7$$
$$\mathrm{Var}(S) = \frac{35}{6}$$

(b) A condition affects 1% of a population. A test detects it in 99% of the people who have it, and comes back positive for 5% of the people who do not. A randomly chosen person tests positive. What is the probability that they have the condition?

Solution.

Let c represent the person having condition
Let p represent the test is positive

$$P(c|p) = \frac{P(p|c)P(c)}{P(p)} = \frac{1}{6}$$

(c) Let $X_1, \ldots, X_n$ be independent with the same distribution, mean μ, and variance σ^2, and let $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Compute $\mathbb{E}[\bar{X}_n]$ and $\mathrm{Var}(\bar{X}_n)$. If $\sigma = 2$, how large must n be for the standard deviation of $\bar{X}_n$ to be at most 0.1?

Solution.

$$\mathbb{E}[\bar{X}_n] = \mu$$

$$\mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$$

$$n \geq 400$$

If this felt unfamiliar: the probability review linked on the course page (expectation, variance, independence, conditional probability).

2 Linear Algebra

(a) Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Compute Ab. Solve $Ax = b$ for $x \in \mathbb{R}^2$. Find the eigenvalues of A and an eigenvector for each. Is A positive definite?

Solution.

$$Ab = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$$

$$Ax = b \iff \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} x = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \Rightarrow x = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$|\lambda E - A| = 0 \Rightarrow \lambda_1 = 1 \quad \& \quad \lambda_2 = 3$$

(a) $\lambda = 1$

$$x = k \begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad \forall k \neq 0$$

(b) $\lambda = 3$

$$x = k \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \forall k \neq 0$$

Since $2 > 0$ and $2 \times 2 - 1 \times 1 > 0$, A is positive definite

(b) Let $X \in \mathbb{R}^{n \times d}$. Show that $X^\top X$ is symmetric and positive semidefinite, and that it is positive definite if and only if the columns of X are linearly independent.

Solution.

$$(X^T X)^T = X^T (X^T)^T = X^T X \Rightarrow X^T X \text{ is symmetric}$$

$$x^T X^T X x = (Xx)^T Xx = \|Xx\|^2 \geq 0 \Rightarrow X^X \text{ is positive semidefinite}$$

(a) $\Rightarrow$
Suppose X^TX positive definite. Let $y = X^TX$ then $\forall y_i, y_i > 0$
Let $s_i = (x_{1i}...x_{ni})^T$ Then we have

$$\Sigma_{i=1}^{d} s_i x_i \neq 0$$

when for $\forall (x_1...x_d) \neq 0 \Rightarrow$ X is linearly independent

(b) $\Leftarrow$
Suppose X linearly independent. Then for $\forall (x_1...x_d)^T \neq 0$. We have $Xx \neq 0 \Rightarrow (Xx)^T Xx > 0 \Rightarrow X^TX$ positive definite

If this felt unfamiliar: the linear algebra review linked on the course page (matrix products, inverses, eigenvalues, positive definiteness).

3 Calculus

(a) Let $f(x, y) = x^2 + xy + y^2 - 3x$. Compute the gradient ∇f and the Hessian $\nabla^2 f$. Find the point where the gradient vanishes, explain why it is the global minimizer of f, and give the minimum value. Finally, write down the point reached by one step of gradient descent from $(0, 0)$ with step size $\eta = 0.1$.

Solution.

$$\nabla f = \begin{pmatrix} 2x + y - 3 \\ x + 2y \end{pmatrix}$$

$$H = \nabla^2 f = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

$$\nabla f = 0 \Rightarrow \begin{cases} 2x + y - 3 = 0 \\ x + 2y = 0 \end{cases} \Rightarrow \begin{cases} x = 2 \\ y = -1 \end{cases}$$

Since $2 > 0$ and $2 \times 2 - 1 \times 1 > 0 \Rightarrow H$ is positive definite $\Rightarrow f$ strictly convex $\Rightarrow (2, -1)$ is the unique minimum and $f(2, -1) = -3$

$$x = \begin{pmatrix} 0 \\ 0 \end{pmatrix} - 0.1 \times \begin{pmatrix} -3 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.3 \\ 0 \end{pmatrix}$$

(b) Let $X \in \mathbb{R}^{n \times d}$ have linearly independent columns, let $y \in \mathbb{R}^n$, and let $f(w) = \frac{1}{2}\|Xw - y\|_2^2$ for $w \in \mathbb{R}^d$. Compute $\nabla f(w)$ and find the minimizer $w^\star$ (you may use Problem 2(b)).

Solution.

$$\nabla f(w) = X^T(Xw - y)$$

$$w^* = (X^TX)^{-1}X^Ty$$

(c) Let $\sigma(z) = 1/(1 + e^{-z})$ for $z \in \mathbb{R}$. Show that $\sigma'(z) = \sigma(z)\big(1 - \sigma(z)\big)$. Then, for a fixed $x \in \mathbb{R}^d$, compute the gradient with respect to $w \in \mathbb{R}^d$ of $g(w) = \sigma(w^\top x)$.

Solution. Let $u = 1 + e^{-z}$.

$$\sigma'(z) = -\frac{1}{u^2} \times u' = -\sigma^2(z) \times (\frac{-1}{\sigma(z)} + 1) = \sigma(z) - \sigma^2(z) = \sigma(z)(1 - \sigma(z))$$

$$\nabla g(w) = \sigma(w^\top x)(1 - \sigma(w^\top x))x$$

If this felt unfamiliar: the matrix calculus part of the same linear algebra review (gradients, Hessians, least squares), and any calculus text for the chain rule.

4 Python and NumPy

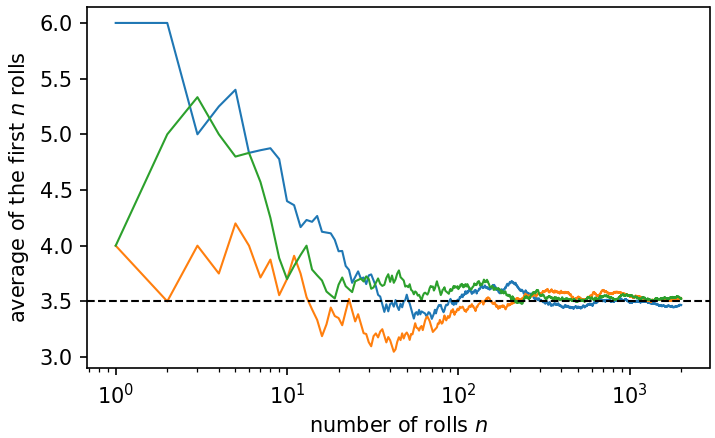
The programming homeworks use Python 3.10 or newer with NumPy and Matplotlib. Set up a Python environment that you will keep for the semester (`venv` or `conda`), install the two packages, and download `hw0.py`, released with this handout.

(a) Complete the two functions marked `TODO` in `hw0.py`: `standardize_columns`, which subtracts from each column of a 2-D array the column's mean and divides by the column's standard deviation, without a Python loop (hint: the `axis` argument and broadcasting); and `least_squares`, which computes the minimizer $w^\star$ of Problem 3(b) (`np.linalg.solve` is preferable to forming an inverse).

(b) Run `python hw0.py`. It prints your Python, NumPy, and Matplotlib versions, checks the two functions against known answers, and saves `hw0_fig.png`: the running average of repeated rolls of a die, which should settle near the $\mathbb{E}[X]$ of Problem 1(a) (nothing to code). Paste the output and include the figure in your PDF; `hw0.py` itself is not submitted.

Solution.

```
[PASS] standardize_columns: [[1, 2], [3, 6]] -> [[-1.0, -1.0], [1.0, 1.0]],
expected [[-1.0, -1.0], [1.0, 1.0]]
[PASS] least_squares: X = [[1, 0], [1, 1], [1, 2]],
y = [1, 3, 5] -> w = [1.0, 2.0], expected [1.0, 2.0]
2 of 2 checks passed.
```



If this felt unfamiliar: the Python/NumPy tutorial linked on the course page, in particular array indexing, broadcasting, and the `axis` argument.

5 Self-Assessment (required)

For each of Problems 1–4, write one line: comfortable, rusty, or new; roughly how long it took; and, if rusty or new, what you will review and by when (for example, "Problem 2: rusty; 25 minutes; eigenvalues and positive definiteness, from the linear algebra review, by the end of next week"). Be honest: it tells us where the class stands and tells you what to review.

Solution.

1. **Problem 1:** 30 mins
2. **Problem 2:** 45 mins
3. **Problem 3:** 60 mins
4. **Problem 4:** 30 mins

Assistance Statement. List every person and resource, including AI tools, that you used beyond the course materials, and for what. Write "none" if none.

Solution.

Problem 1: None

Problem 2: None

Problem 3: Hessian Matrix and multivariables derivative chain rule

Problem 4: Search the usage of np.linalg

Optional. Anything else you would like the instructor to know (background, concerns, hopes for the course).